# DrainWatch: A Low-Cost Sensor-Integrated Belt to Improve Safety and Patient Mobility During JP Drain Use

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**Abstract**

The Jackson-Pratt (JP) drain is widely used to prevent seroma and infection after breast cancer and reconstructive surgeries, yet patients continue to face challenges such as manual monitoring, suction loss, discomfort, and restricted mobility. Prior studies, including Chen et al. (2016), Sakkary (2012), and Gadiraju et al. (2025), show that prolonged drain use and inconsistent monitoring contribute to higher complication rates, emphasizing the need for more user-friendly postoperative solutions. This project explores opportunities to modernize JP drain management through a wearable belt designed to improve comfort and provide basic, real-time feedback. The proposed system integrates a simple, non-contact sensor, such as a weight-based volume estimator or external pressure sensor, within a soft, washable belt that stabilizes the drain bulb and reduces tugging. By combining ergonomic support with accessible, low-power monitoring, this design aims to enhance safety, ease of use, and patient confidence during at-home recovery.

## Dedication

This report is dedicated to postoperative patients recovering from breast cancer, reconstructive surgeries, and other surgical procedures requiring JP drainage. Your resilience continues to motivate improvements in patient-centered care.


## Acknowledgements

We would like to express our sincere gratitude to the mentors and advisors of the UMEED program, whose insight, encouragement, and early guidance inspired the direction of this project and helped shape the development of DrainWatch. Their expertise was invaluable in refining our ideas and pushing us toward meaningful, patient-centered innovation.

We also thank our course instructors and the ABE faculty for providing resources, feedback, and support throughout the design process. Their instruction gave us the technical foundation needed to bring our concept to life.

Finally, we extend our appreciation to the students, peers, and staff who contributed feedback, which assisted with our designing of the device.

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## INTRODUCTION

Postoperative fluid accumulation is a common complication following breast cancer surgeries, mastectomies, and reconstructive procedures, often leading to seroma, infection, discomfort, and longer recovery times. To manage these risks, the Jackson-Pratt (JP) drain has served as the standard closed-suction device for decades. However, its effectiveness depends heavily on manual monitoring, leaving patients responsible for checking fluid levels, maintaining suction, and recognizing early signs of problems. For many recovering at home, this process can be confusing, inconsistent, and physically difficult, resulting in unnoticed suction loss, poorly recorded volumes, and increased anxiety.

Recent studies highlight that inadequate monitoring and prolonged drain duration significantly contribute to postoperative complications, underscoring the need for more reliable, patient-friendly support. Patients frequently report discomfort from the dangling bulb, frustration with managing the tubing, and uncertainty about whether the drain is functioning properly. Meanwhile, clinicians may receive incomplete information between follow-up visits, challenging their ability to determine when drains can be safely removed.

To address these challenges, our team developed DrainWatch, a simplified, wearable belt system designed to improve comfort and provide basic, real-time feedback on JP drain performance. The device focuses on two achievable and clinically meaningful features for an undergraduate project: stabilizing the bulb with an ergonomic belt to reduce discomfort and integrating a single non-contact sensor, either a load cell for volume estimation or an external pressure sensor to monitor suction integrity. By enhancing ease of use without modifying the drain itself, DrainWatch aims to support patients during recovery and offer clinicians more consistent information, all within a practical and feasible design scope.

## LITERATURE REVIEW / BACKGROUND

### *2.1 General Background*

Postoperative fluid accumulation is one of the most common complications following oncologic, reconstructive, and orthopedic surgeries. Excess serous or lymphatic fluid can cause pain, infection, and delayed wound healing if not properly managed. To prevent these outcomes, closed-suction drainage systems, most notably the Jackson-Pratt (JP) drain, have become standard in surgical practice since their introduction in the 1970s (Bhattacharjee, Ramteke, & Lamture, 2020; Durai & Ng, 2010). The JP drain consists of a perforated silicone tube attached to a compressible reservoir bulb that provides gentle negative pressure to evacuate wound exudate. This simple and effective design has led to widespread adoption across multiple disciplines, including general, orthopedic, and reconstructive surgery (Bartosz et al., 2021;

Ravikumar et al., 2001). Figure 1 shows the structure of Jackson-Pratt and its conventional placement on the body.

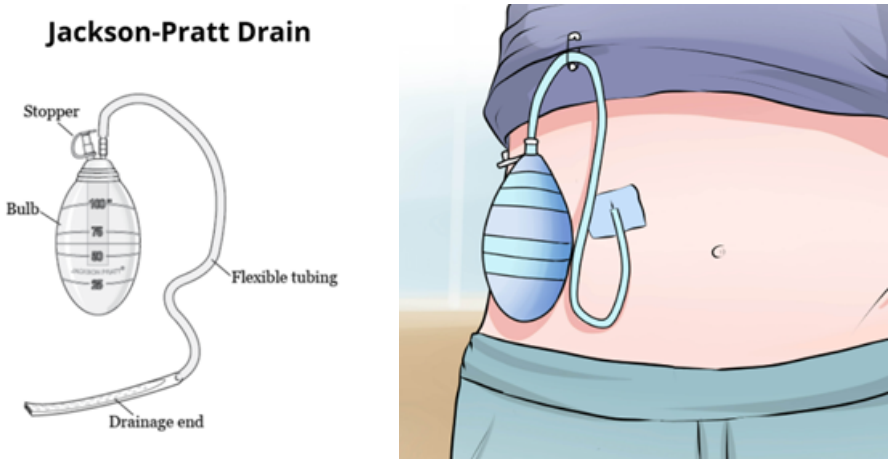


**Figure 1.** *Illustration of a Jackson-Pratt drain and its conventional placement on the body following surgery* (Saint John's Cancer Institute, n.d.; WikiHow, n.d.).

Among its most frequent indications is breast cancer surgery, where tissue dissection and lymph node removal create large dead spaces prone to seroma formation. Studies report that 60–80 percent of patients undergoing mastectomy or immediate breast reconstruction require at least one JP drain postoperatively (Chen, Lin, Hung, & Chou, 2016). Globally, with more than 2.3 million new breast cancer cases each year (World Health Organization, 2024), this translates into millions of drains placed annually. Despite their ubiquity, JP drains present persistent drawbacks that compromise patient comfort and recovery. Chen et al. (2016) found that drain duration, not output volume, correlated most strongly with surgical-site infection, while Sakkary (2012) showed that excessive drainage time increased seroma risk after mastectomy. Gadiraju et al. (2025) further demonstrated that prolonged drain use significantly decreases quality of life, with patients reporting limited mobility, anxiety, and discomfort from accidental tugs and leakage.

Clinically, these findings expose a crucial gap: current JP management relies on manual, subjective monitoring of drainage volume and suction pressure. Patients must empty and measure the bulb periodically, record volumes, and identify early infection signs through visual inspection. In reality, compliance is inconsistent and dependent on education, lighting, and dexterity. Even in hospital settings, suction failure or tubing occlusion may go unnoticed until seroma or hematoma develops (Durai & Mownah, 2009). Furthermore, the reservoir's vacuum often degrades over time, reducing drainage efficacy, while infection or blockage within the tubing may present only subtle changes in fluid color or odor parameters difficult to quantify without instrumentation.

Parallel to these challenges, recent advances in wearable biosensing and smart textiles have revolutionized real-time physiological monitoring. Flexible and biocompatible micro-sensors now enable detection of temperature, pH, turbidity, flow, and pressure in compact, low-power

packages suitable for integration into clothing or soft supports (Lee et al., 2023; Zhao et al., 2022). Biomedical engineers have begun embedding similar sensors into wound dressings and negative-pressure therapy systems to provide continuous feedback on healing conditions and device performance. These technological trends open the door to a next-generation JP system that couples traditional suction drainage with intelligent monitoring and ergonomic design.

A promising concept emerging from this convergence is a smart wearable belt capable of housing and supporting the JP reservoir while hosting embedded sensors. In such a configuration, ultrasonic or weight sensors could track fluid volume; MEMS pressure transducers could ensure consistent suction; optical or electrochemical modules could assess turbidity, pH, and temperature for early infection detection; and micro-flow or acoustic sensors could detect tubing blockage. All signals could be processed by a miniature, battery-powered controller sewn into the belt, transmitting alerts wirelessly to clinicians or mobile devices. Beyond monitoring, the belt's ergonomic form, constructed from antimicrobial, elastic fabrics, would secure the bulb comfortably against the body, prevent accidental pulling, and restore mobility and confidence during recovery.

Collectively, these advancements reflect an opportunity to transform the JP drain from a passive device into an active, data-driven system that supports both clinicians and patients. By integrating sensing, comfort, and communication, such innovation could reduce infection rates, shorten hospital stays, and enhance quality of life for breast-cancer survivors and other postoperative patients alike.

*2.2 Specific Dilemma*

While the potential for a smart JP system is compelling, several obstacles underline the need for targeted research and design. The current drain remains indispensable in surgery, yet its lack of objective monitoring leads to either premature removal, risking seroma, or prolonged retention, risking infection. Introducing electronic components into a fluid-filled, sterile system poses challenges in biocompatibility, waterproofing, sterilization, and regulatory approval. Additionally, any wearable integration must balance sensor proximity to the drain with patient comfort and hygiene, ensuring that electronics never compromise sterility or add weight that strains the incision site. Power management, data security, and false-alarm minimization also demand consideration.

These tensions define the central dilemma: how to modernize the Jackson-Pratt drain with intelligent sensing and ergonomic support while preserving its safety, simplicity, and affordability. Addressing this gap requires a multidisciplinary approach that merges surgical insight, materials science, and biomedical electronics to design a next-generation drainage solution capable of improving postoperative outcomes for breast-cancer and other surgical patients worldwide.

*2.3 Other Research*

Contemporary research on Jackson-Pratt and other closed-suction drains reflects a growing shift from purely mechanical drainage toward smart, data-enabled systems designed to improve safety, efficiency, and patient comfort. One major focus area is the automation of drain maintenance. Seol et al. (2023) developed an automatic Jackson-Pratt auxiliary device that mechanically "strips" the tubing to remove clots and maintain suction without manual manipulation, achieving a 93.8% stripping efficiency in experimental trials. Such innovations aim to reduce nursing workload and prevent clogging, one of the most frequent causes of drain failure. However, while effective at preserving flow, these systems remain largely mechanical and lack real-time monitoring capabilities.

Another active domain of research involves the integration of sensors for quantitative fluid monitoring. Roser et al. (2021) introduced a Smart Sensor System that continuously records drainage flow dynamics and composition, demonstrating reliable, real-time characterization of postoperative fluid behavior. Similarly, Choudhury et al. (2021) created a biosensing prototype that automated postoperative drain monitoring in a peritoneal injury model, achieving an $R^2 = 0.9976$ correlation with manual flow measurements, illustrating the feasibility of accurate, continuous fluid quantification. Extending this concept further, Meckler et al. (2025) embedded a mini spectrometer into a surgical drainage system to analyze biochemical markers such as hemoglobin, bilirubin, and lactate dehydrogenase directly within the drain output. Their system achieved strong agreement with laboratory assays (AUC > 0.9 across analytes), representing an important step toward diagnostic drainage monitoring capable of distinguishing seroma, hemorrhage, and infection in real time.

Parallel work has explored computer-vision–based drain assessment as a low-cost alternative to embedded sensors. Sahin et al. (2024) employed deep learning and semantic segmentation to estimate fluid volume from photographs of JP bulbs, achieving high accuracy across more than 1,000 labeled images. This approach simplifies volume recording in outpatient care but still depends on patient participation and lacks autonomous data collection. Other comparative and clinical studies continue to evaluate drain design and outcomes. Lattoo et al. (2024) compared Jackson-Pratt drains with passive gravity drains following major abdominal surgery, reporting lower rates of clogging, reduced pain, and fewer interventions in the JP cohort, confirming its continued clinical superiority. Meanwhile, ongoing clinical trials, such as the Interi System study (National Cancer Institute, 2024), are testing alternative manifold-based drainage architectures in post-mastectomy and reconstructive breast cancer patients to reduce seroma formation and shorten drainage duration.

Earlier efforts to digitize drainage have also appeared in the literature and patent landscape. van Duren et al. (2019) described a prototype using conductive electrodes to electronically detect fluid levels and trigger alarms, reducing manual measurements, while U.S. Patent 7,267,671 B2 proposed incorporating optical reflectance sensors into surgical drains for continuous monitoring. More recently, multidisciplinary research in wearable biosensing has introduced textile-integrated pH, temperature, and flow sensors that could readily translate into next-

generation surgical drains (Zhao et al., 2022; Lee et al., 2023). Collectively, these advances illustrate an accelerating effort to modernize drainage management through automation, sensing, and data connectivity.

Despite these promising developments, current research remains fragmented; each project addresses only a single failure mode, such as blockage, fluid quantification, or biochemical analysis. None yet combines multiple sensor modalities within a wearable, patient-centered platform capable of monitoring suction integrity, infection biomarkers, and fluid flow simultaneously. The proposed Jackson-Pratt smart belt seeks to bridge this gap by integrating MEMS pressure, ultrasonic or weight-based level sensing, turbidity, pH, and flow modules into flexible, washable textile system. By merging the mechanical reliability of the traditional JP drain with the intelligence of modern biosensing and the comfort of ergonomic design, this project extends ongoing research toward a fully integrated, real-time postoperative monitoring solution for breast cancer and other surgical patients.

### *2.4 Market Analysis*

#### *2.4.1 Market Size*

The global market for surgical drainage devices, including systems like the Jackson-Pratt drain, was valued at approximately USD 2.47 billion in 2022 with a projected compound annual growth rate (CAGR) of 6.4% from 2023 to 2030, reaching about USD 4.04 billion by 2030 (Grand View Research, 2024). Another projection suggested a value of USD 3.0 billion by 2024 and USD 5.3 billion by 2033 at 6.6% CAGR (Dimension Market Research, 2025). These figures confirm robust demand driven by rising surgical volumes and a focus on postoperative outcomes.

#### *2.4.2 Existing Technologies*

Currently, most drainage systems remain passive or mechanical, lacking digital monitoring. Recent prototypes integrate sensors for flow, suction, and biochemical analysis (Roser et al. 2022; Meckler et al., 2025), while wearable biosensing textiles (Lee et al., 2023; Zhao et al., 2022) demonstrate feasibility for flexible integration. Yet, a fully connected, multi-sensor wearable JP system remains unrealized.

#### *2.4.3 Stakeholders*

The development and implementation of a smart Jackson-Pratt drainage system involves a wide network of stakeholders whose needs and perspectives shape both design priorities and real-world adoption. Understanding their unique roles is crucial for aligning technical innovation with clinical practicality and user acceptance. Each stakeholder group, ranging from patients and clinicians to manufacturers and regulatory bodies, contributes to the success of the device by influencing usability, compliance, safety, and market viability.

- Patients: desire comfort, mobility, and safety.
- Clinicians: require accurate data and early alerts.
- Hospitals: seek reduced infection rates and shorter stays.
- Manufacturers: pursue differentiation through innovation.
- Insurers: prioritize devices that reduce readmission costs.
- Regulators: ensure safety, biocompatibility, and data protection.

*2.4.4 Regulations*

JP drains are Class II medical devices under FDA 21 CFR 878, requiring 510(k) clearance (Registrar Corp, 2025). Integrating sensors adds layers of regulation:

- Biocompatibility (ISO 10993-1) for patient-contacting materials (FDA, 2024).
- Sterility and reprocessing validation
- Electrical safety and EMC compliance.
- Cybersecurity and data integrity for connected components.

Regulatory strategies must address both sterile medical-device and wearable electronics standards.

*2.5 Conclusions*

The Jackson-Pratt drain has served surgery for decades but remains limited by manual monitoring and discomfort. The proposed smart, wearable JP system offers a feasible, data-driven evolution integrating biosensors for suction, flow, pH, temperature, and infection detection within an ergonomic textile belt. Grounded in recent biosensor and wearable advances (Lee et al., 2023; Zhao et al., 2022), it can transform postoperative recovery by improving comfort autonomy, and safety while supporting clinicians with continuous data. Future directions include prototyping, clinical validation, and regulatory alignment to ensure safety, reliability, and manufacturability. With multidisciplinary collaboration, the smart JP drain represents the next frontier in patient-centered, technology-enhanced surgical care.

## CONCEPT GENERATION & EVALUATION

*3.1 Concept Generation*

The first objective in the generation of a concept was the overall core goal of developing a wearable system that comfortably holds JP drain bulbs for the two main surgeries types they are used in, and to contain one or two basic, low-risk, non-contact sensors that provide useful monitoring without interacting with bodily fluid. Conceptualization first started in brainstorming

groups, where ideas of any size were stated and written down. This first stage did not include evaluating whether the ideas were feasible yet. Ideas were even built off of each other, everyone taking inspiration from one another until everyone was satisfied with the amount of ideas there were to further evaluate. Our first proposal was a conglomeration of all the ideas: a comprehensive, fully integrated smart drainage belt with multi-modal sensors (pH, temperature, turbidity, flow, pressure, spectrometry), wireless connectivity, algorithms, and a fully washable wearable system. The first idea (seen in the figures below) for the physical design was that it needed to be made in a way that could allow for holding of the bulbs after a mastectomy or hip surgery, and that could be able to be washed for sterilization. The washable aspect is useful for when small droplets of the bodily fluid drip onto the fabric.

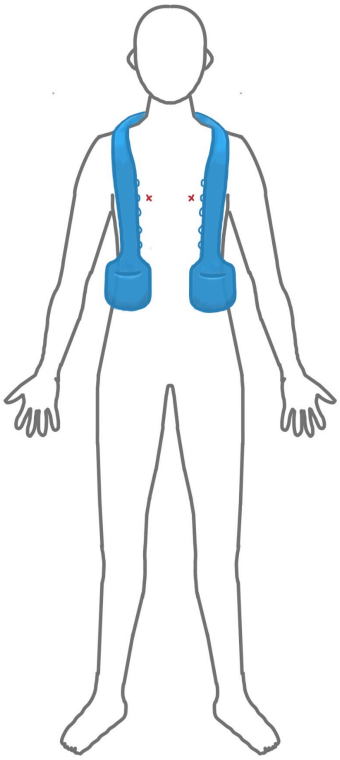

**Figure 2.** *Rough sketch of the first design idea for the bulb holder in case of a mastectomy. The red x's are the incision points. The bulbs would go into the pouches.*

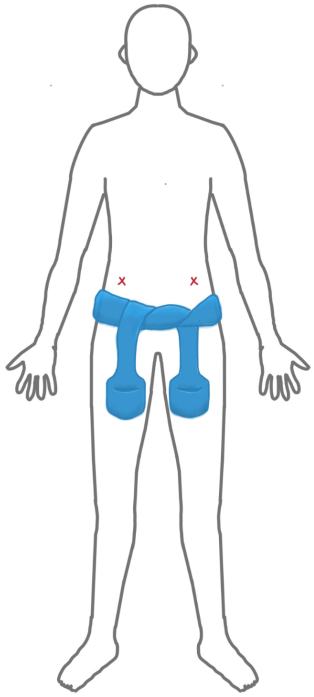

**Figure 2.** *Rough sketch of the first design idea for the bulb holder in case of a hip surgery. The device is the same, and the sketch shows it being worn in a versatile way to work as a holder for multple surgeries. The red x's are the incision points. The bulbs would go into the pouches.*

General filtration of ideas was based on the physical design already established and feasibility of achievement. The first large concept that was removed was the idea of the smart sensors being able to wirelessly send the information detected by sensors to an app or clinic portal viewable by doctors or the patient. While a simple app or interface may still be achievable, the patient portal may be too costly, and would have to follow various requirements upheld by healthcare regulation and privacy laws when dealing with patient's data being stored online. This would make the development of a portal with the proper software too complex for the project. However, a simple interface may still be able to be incorporated, so long as it does not have to store any private information. The multi-sensoring biochemical monitoring was decided to be unfeasible second. Even though this component would allow for pH, turbidity, temperature, and hydration assessment through spectrometry and in-fluid sensing (all of which are wonderful statistics to have for a patient to monitor health), the idea would require something to be inside of the tubing, not on the holder. This would then deal with the reconstruction of the JP tubing and bulbs in a costly manner that would overall be too difficult due to upholding of FDA regulations, being able to actually test if monitors worked, too costly, and risking infection from more items being exposed to bodily fluids inside the tubing. Other monitors, such as the clog detection monitors and flow sensors, were also deemed unfeasible for the same reason.

*3.2 Concept Evaluation*

The evaluations went through two processes: first the design was decided, then the ideas were further filtered to fit the design. The first evaluation was done for the physical concept of the design. The general concept was a versatile device that could be used on two separate parts of the body depending on whether the patient underwent a mastectomy or hip surgery. However, this design was simplified to allow the group to focus more on the sensor aspect, as well as allow for more space to put the sensors. The device is now just a belt with specific placement around the waist to hold bulbs the same way, no matter which surgery the patient underwent. The evaluation was done on the ideas left over after filtering on how to make the ideas fit the functionality and design of the holder. Once the general design was decided and sketched (Figure 3), the second evaluation was done through making requirements that must be meet in order for the holder to meet the main goal. The requirements were: fabric of belt must be lightweight to not put a strain on the patient, the fabric must be washable, the belt must feature a way of holding the bulbs in a secure nature, and the sensor included must be removable for when fabric is being washed.

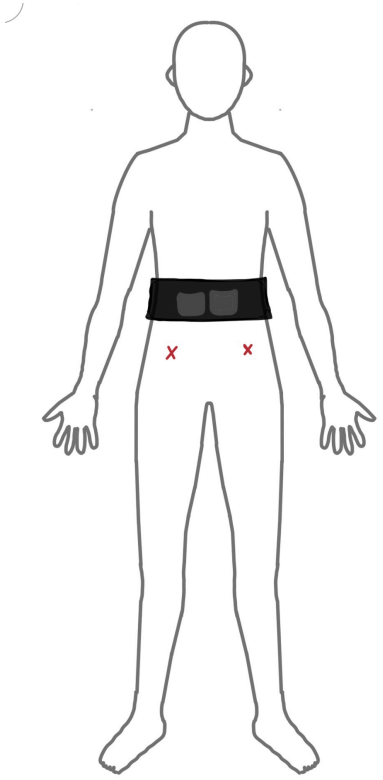

**Figure 3.** *Rough sketch of the second design idea for the bulb holder in case of either surgery The red x's are the incision points for hip surgery, and the red x's for a mastectomy are not included simply due to the clear distance from the chest. The bulbs would go into the pouches (highlighted gray spots.)*

Our choices for materials mainly focused on fabrics due to the fact that the belt should be machine washable for sterilization. So, narrowing the material to fabrics, the next essential to met is the fact the material must be lightweight do that the belt is not to heavy for a patient recovering from a surgery. Through a research of multiple lightweight materials that are machine washable, the best choice seemed to be neoprene. The quality that stood out the most that sold the group on the idea of neoprene is that it is used for gear where a durable material is needed, but mobility and comfort is also important. This perfectly fit our requirements for the belt material. The next aspect of the belt that was important is the way that the bulbs would be held up in the belt. The way the bulbs would need to be held would be in a way that cups the bulbs, providing support from all around in order to keep the bulb from dangling and pulling on the patient. The idea that was decided on was holsters on the outside of the belt that are, as of right now, made of the same material as the belt itself. As for sensor selection, the remaining sensor options were for weight-based volume estimation, pressure sensor to track suction loss, or a temperature sensor on belt for indirect infection indication. To simplify the project and make the belt a lighter weight, the group decided to go with only one sensor. Since the problem with the other sensors were that they would have to be inside the bulb, the primary want for this sensor was the one that could get the most information with the least amount of interference with the bulb. A temperature sensor may not always be an accurate measurement for an infection since they may occur without heating at incision site. Suction loss can be visible seen when the fluids stop flowing in the tub. So, the sensor decided on was the weight-based volume estimation sensor, which is explained in detail in the final design section. The weight-based volume estimation sensor also works well with the holster design already chosen, as it can just be used as a load cell put into the bottom of the holster.

### *3.3 Expert Feedback*

For feedback on the concept, Laura Geter was asked her opinion. Laura Geter has been a registered nurse (RN) for 27 years, has a Bachelor of Science degree in nursing, and has had Jackson Pratt drainage tubes put in after a mastectomy due to breast cancer. She has been on both the nurse side of caring for and maintaining the tubes, as well as the patient side. Laura's response was that she thinks this device would be very helpful for a patient to have a place to secure their JP bulb. She has seen many complications in which simply using medical tape to tape the bulbs to the patient side causes an adverse reaction on the patient. The fact that it could be worn comfortable under loose clothing was another aspect she liked. This aspect promotes patient dignity, which is a huge idea in the medical field. In her opinion, she also believed the system would be useful for alerting nursing staff when the bulbs are full and would decrease the chance of postoperative problems, including patient pain. She explained that in a more acute care setting, such as a hospital, if the patient is putting out too much drainage too rapidly, it would alert the nursing staff to empty out the collection bulb. If the bulb is full, then fluid is collecting inside the body and suction stops. The fluid build-up will risk infection, cause inflammation, and result in patient pain. There will be an increased susceptibility to postoperative complications.

Going into the final design, the group's chosen concept is a lightweight belt that functions as a JP drainage bulb holder that can tell the user the volume or "fullness" of bodily fluids drained into the bulb in real time. This design is the best one by far, not only due to the potential of easing a patient's pain, but also due to the fact that all of the components fit together so well in this concept. The concept incorporates all that it needs to, and can still function in the way it needs as well.

## FINAL DESIGN AND APPROACH

### *4.1 Final Design & Approach*

The final design of DrainWatch focuses on creating a wearable system that stabilizes the JP drain bulb and incorporates one simple, non-contact sensor to assist with basic postoperative monitoring. Our approach emphasizes feasibility, patient comfort, and low manufacturing complexity, aligning with the scope of a single-semester project. The design integrates three core components: the belt structure, the bulb holster, and the sensor subsystem with supporting electronics. Together, these elements create a system that improves mobility, reduces tubing tension, and provides real-time feedback on drain function without introducing sterility or regulatory concerns.

### *4.2 Belt Structure*

The belt serves as the primary support system and is designed to comfortably secure the JP drain bulb during daily activities. The belt is made from a soft, flexible material such as elastic or neoprene to allow a snug but comfortable fit around the torso. This material selection reduces irritation and prevents shifting during movement. The belt includes adjustable Velcro closures to accommodate a range of patient body sizes and to allow easy tightening or loosening with one hand if needed.

A key focus of this design was ensuring that the belt could be worn under clothing without discomfort. The belt width was chosen to distribute pressure evenly and reduce friction against the skin. The interior surfaces are smooth, and all edges are finished to prevent abrasion during extended wear. Because sterility and cleaning were concerns identified early, the electronics are housed in a removable module so the belt itself remains fully machine-washable.

### *4.3 JP Bulb Holster Design*

The holster secures the drain bulb and prevents it from dangling, which was a major patient pain point identified during background research. The holster is positioned at the front or side of the belt depending on patient preference. It is constructed using either fabric reinforced with padding

or a 3D-printed insert with a soft silicone lining. Both configurations were explored to ensure stability while protecting the bulb from compression or accidental impact.

The holster is shaped to match the geometry of a standard Jackson-Pratt bulb, allowing it to sit upright and preventing rotation that could twist the tubing. Openings at the top allow the tubing to pass through without kinking. The holster placement also keeps the sensor in consistent contact with the base of the bulb when using the load cell configuration or maintains steady alignment against the bulb surface if the pressure sensor option is used. This positioning supports more reliable data collection while enabling ease of insertion and removal of the bulb during emptying.

### *4.4 Sensor Selection and Integration*

Because the simplified project plan recommended selecting only one feasible sensor, our team chose the load cell option for weight-based volume estimation. This sensor allows the system to approximate reservoir fullness without contacting bodily fluid. The load cell is placed beneath the bulb inside the holster, with a small rigid platform distributing the bulb weight evenly across the sensor surface. The load cell is paired with an HX711 amplifier to improve resolution and stability in readings.

This approach was selected due to its simplicity, low component cost, accuracy under static loads, and minimal electrical risk. The sensor readings provide a basic indication of whether the bulb is approaching capacity, which can inform the patient when the drain should be emptied. While not clinical-grade, the signal provides valuable trend data within the intended educational and proof-of-concept scope.

Future modifications could allow swapping in an external pressure sensor using the same housing and microcontroller structure. That option would enable monitoring of suction loss, another major concern during recovery, but was not implemented in the current build due to the narrowed scope.

### *4.5 Electronics and Microcontroller System*

The electronics subsystem is centered around an Arduino or ESP32 microcontroller, chosen for ease of integration, reliability, and compatibility with the HX711 amplifier. The microcontroller is enclosed in a small 3D-printed housing that mounts to the exterior of the belt. This enclosure protects the electronics from moisture and provides a secure place to route the sensor wiring.

The system is powered by a rechargeable battery connected through a simple on-off switch. Because this design avoids features requiring high power consumption such as continuous

wireless communication, the battery life remains adequate for daily use. During operation, the microcontroller collects load cell data, filters noise, and converts the signal into an estimated volume level. The processed values are displayed through a laptop serial monitor or a basic optional mobile interface that demonstrates proof-of-concept visualization.

### *4.6 Overall System Operation*

To use the system, the patient secures the belt around the waist and places the JP bulb inside the holster. As the bulb collects fluid, its weight changes, and the load cell detects the corresponding force. The microcontroller processes this data and outputs a real-time volume estimate or fullness percentage. The design stabilizes the bulb, reduces tubing strain, and provides simple feedback without requiring any modification to the drain itself.

By focusing on comfort, simplicity, and achievable engineering solutions, the DrainWatch system meets the goals outlined at the beginning of the project. The final design supports patient mobility and postoperative confidence while remaining feasible for undergraduate development and testing.

## DISCUSSION & CONCLUSIONS

### *5.1 Discussion*

The development of DrainWatch illustrates how a focused engineering approach can address persistent shortcomings in postoperative Jackon-Pratt (JP) drain management. The literature identifies several key challenges: manual volume tracking, suction uncertainty, patient discomfort, and inconsistent at-home monitoring, that directly informed the design decisions made throughout this project.

The design decisions centered on feasibility, focusing on engineering features that could be realistically achieved with a single-semester project. Early brainstorming explored multiple sensing modalities described in current research, including pressure transducers, flow sensors, turbidity indicators, and biochemical markers. However, regulatory constraints, sterility concerns, and system complexity made most options unsuitable for the current scope. This evaluation process led to the selection of a single non-contact load cell sensor to estimate bulb fullness based on reservoir weight. The sensor avoids sterility barriers, is inexpensive, and aligns with patient and clinician needs for clearer volume assessment.

The wearable belt and holster design also emerged from user-centered considerations. Literature showed that patients frequently struggle with the dangling bulb, accidental tugs, and discomfort caused by unstable tubing. By integrating the JP bulb into a soft, ergonomic belt structure, DrainWatch aims to improve mobility and reduce strain on the tubing. The removable electronics

housing addresses hygiene and safety by allowing the fabric components to be machine-washed while keeping circuitry isolated from moisture and bodily fluids.

Throughout the design process, ethical and societal considerations were incorporated. The team prioritized patient safety, accessibility, and simplicity to create a device that could be adopted across diverse clinical and socioeconomic settings. Cost, manufacturability, and power requirements were also evaluated to ensure that the concept remains realistic for long-term development beyond this course. The project required independent research across biomedical engineering, wearable design, and sensor integration, allowing the team to critically evaluate alternative concepts and converge on a practical, impactful solution.

In its current form, DrainWatch provides a promising direction for improving patient comfort, reducing anxiety, and supporting more consistent drain monitoring during recovery. Future work will include constructing a physical prototype, validating sensor accuracy with simulated fluid, assessing user experience with preliminary trials, and exploring additional monitoring features. With continued refinement and testing, DrainWatch has the potential to evolve into an accessible and patient-centered solution that meaningfully improves postoperative care.

## REFERENCES


Bartosz, P., Grzelecki, D., Chaberek, S., Para, M., Marczyński, W., & et al. (2022). A prospective randomized study: Use of closed suction drainage after revision hip arthroplasty may lead to excessive blood loss. *Scientific Reports, 12*, 881. https://doi.org/10.1038/s41598-022-05023-2

Bartosz, P., Marczyński, W., Para, M., Kogut, M., & Białecki, J. (2021). Comparative study of suction drainage placement in cementless hip replacement among patients undergoing extended thromboprophylaxis: A prospective randomized study. *BMC Musculoskeletal Disorders, 22*, 688. https://doi.org/10.1186/s12891-021-04583-0

Bhattacharjee, A. A., Ramteke, H., & Lamture, Y. R. (2020). Surgical drains – a review. *Journal of Evolution of Medical and Dental Sciences, 9*(52), 3992–3997. https://doi.org/10.14260/jemds/2020/872

Cardinal Health. (n.d.). *Jackson-Pratt™ channel drains.* Cardinal Health Product Catalog. Retrieved September 2025, from https://www.cardinalhealth.com/en/product-solutions/medical/surgical-products/wound-drainage/jackson-pratt-drains.html

Chen, C.-F., Lin, S.-F., Hung, C.-F., & Chou, P. (2016). Risk of infection is associated more with drain duration than daily drainage volume in prosthesis-based breast reconstruction: A cohort study. *Medicine (Baltimore), 95*(49), e5605. https://doi.org/10.1097/MD.0000000000005605

Choudhury, S., et al. (2021). Automation of postoperative surgical drain monitoring with novel biosensing technology: Proof of concept in a peritoneal injury model. *ResearchGate.* https://www.researchgate.net/publication/348146753

Dimension Market Research. (2025). *Surgical drainage devices market is expected to reach a revenue of USD 5.3 Bn by 2033 at 6.6% CAGR. GlobeNewswire.* https://www.globenewswire.com/news-release/2025/01/28/3016550/0/en/Surgical-Drainage-Devices-Market-Is-Expected-To-Reach-A-Revenue-Of-USD-5-3-Bn-By-2033-At-6-6-CAGR-Dimension-Market-Research.html

Durai, R., Mownah, A., & Ng, P. C. H. (2009). Use of drains in surgery: A review. *Journal of Perioperative Practice, 19*(6), 180–186. https://doi.org/10.1177/175045890901900603

Durai, R., & Ng, P. C. H. (2010). Surgical vacuum drains: Types, uses, and complications. *AORN Journal, 91*(2), 266–274. https://doi.org/10.1016/j.aorn.2009.09.024

Electronic Code of Federal Regulations (eCFR). (2024). *21 CFR Part 878 — General and Plastic Surgery Devices.* U.S. Government Publishing Office. https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-878

Gadiraju, G. K., Tanna, N., De Souza, S., Nizam, S., Brown, R., & et al. (2025). Understanding the impacts of surgical drains on postoperative pain and quality of life. *Plastic and Reconstructive Surgery – Global Open, 13*(1), e6474. https://doi.org/10.1097/GOX.0000000000006474

Grand View Research. (2024). *Surgical Drainage Devices Market Size, Share & Trends Analysis Report by Product (Active, Passive), by Application (Orthopedic Surgery, Cardiac & Thoracic Surgery), by Region, and Segment Forecasts, 2023–2030.* Grand View Research. https://www.grandviewresearch.com/industry-analysis/surgical-drainage-devices-market

Hartanto, V. H., Han, K. R., Ankem, M., & Diamond, S. M. (2001). Endoscopic retrieval of retained Jackson-Pratt drain. *Urology, 57*(5), 973–974. https://doi.org/10.1016/S0090-4295(00)01103-1

Heller, L., Makonnen, R., Zarneke, O., & Bah, A. (2022). *Improved method of securing surgical drains.* University of Wisconsin Biomedical Engineering Design Project Report. https://bmedesign.engr.wisc.edu/projects/f22/drain_pain/file/view/c70d8b9b-70ee-4108-9263-3a0f419acd7a/Drain_Pain-Preliminary_Report.pdf

Lattoo, S. A., et al. (2024). A comparative analysis on the utility of closed suction (Jackson-Pratt) versus gravity drains after major abdominal surgery. *Journal of Current Surgery, 21*(3), 370–377.

Lee, H., Zhao, D., & Kim, J. (2023). Wearable and flexible biosensors for continuous health monitoring: Recent advances and future perspectives. *Biosensors, 13*(6), 630. https://doi.org/10.3390/bios13060630

McKesson Medical-Surgical. (n.d.). *Jackson-Pratt wound drain tubing.* Retrieved September 2025, from https://mms.mckesson.com/product/969879/Cardinal-JP-2190

MDPI Sensors. (2019). *Smart wearable belt system with embedded sensors for health monitoring* [Figure]. In *Sensors*, 19(4600). MDPI. https://www.mdpi.com/sensors/sensors-19-04600/article_deploy/html/images/sensors-19-04600-g001.png

Meckler, S., et al. (2025). Point-of-care biochemical characterization of surgical drain output using an integrated mini-spectrometer system. *PLOS ONE, 20*(3), e0325072. https://doi.org/10.1371/journal.pone.0325072

National Cancer Institute. (2024). *Interi system versus traditional drains in bilateral mastectomy with reconstruction [Clinical trial NCI-2024-02053].* https://www.cancer.gov/research/participate/clinical-trials-search/v?id=NCI-2024-02053

Registrar Corp. (2025). *FDA Class II Medical Devices: Registration, Listing, and 510(k) Requirements.* Registrar Corp. https://www.registrarcorp.com/blog/medical-devices/medical-device-registration/fda-class-ii-medical-devices

Ravikumar, K. J., Alwan, T., Fordyce, M. J. F., & Tuson, K. W. R. (2001). Drainage versus non-drainage in total hip arthroplasty: A prospective randomised study. *Hip International, 11*(1), 49–54. https://doi.org/10.1177/112070000101100106

Roser, M., et al. (2021). Smart sensor system for characterization of real-time flow and drainage features in surgical drains. *Scientific Reports, 11*, 9227954. https://doi.org/10.1038/s41598-021-92279-54

Sahin, T., et al. (2024). Deep learning-based fluid volume estimation from surgical drain images. *European Review for Medical and Pharmacological Sciences, 28*(12), 3702–3710.

Saint John's Cancer Institute. (n.d.). *Jackson-Pratt (JP) drain.* Retrieved October 2025, from https://www.saintjohnscancer.org/melanoma/patient-resources/jackson-pratt-jp-drain/

Sakkary, M. A. (2012). The value of mastectomy flap fixation in reducing fluid drainage and seroma formation in breast cancer patients. *World Journal of Surgical Oncology, 10*, Article 8. https://doi.org/10.1186/1477-7819-10-8

Seol, H., et al. (2023). Development of an automatic Jackson-Pratt drain auxiliary system for mechanical stripping. *Sensors and Actuators B: Chemical, 383*, 133523. https://doi.org/10.1016/j.snb.2023.133523

UCSF Health. (n.d.). *Caring for your Jackson-Pratt drain after breast surgery.* Retrieved September 2025, from https://cancer.ucsf.edu/breastcarecenter/treatment/surgical_oncology/jackson_pratt

U.S. Food and Drug Administration (FDA). (2024). *Biocompatibility Assessment Resource Center.* U.S. Department of Health and Human Services. https://www.fda.gov/medical-devices/premarket-submissions-selecting-and-preparing-correct-submission/biocompatibility-assessment-resource-center

Van Duren, S., et al. (2019). A novel method for electronic measurement and recording of surgical drain output. *Medical Engineering & Physics, 69*, 35–42. https://doi.org/10.1016/j.medengphy.2019.05.004

WikiHow. (n.d.). *How to care for (Jackson-Pratt) JP drains.* Retrieved October 2025, from https://www.wikihow.com/Care-for-%28Jackson-Pratt%29-JP-Drains

World Health Organization. (2024). *Global breast cancer statistics [Fact sheet].* https://www.who.int/news-room/fact-sheets/detail/breast-cancer

Zhao, D., Lee, H., & Zhang, X. (2022). Flexible biocompatible sensors for wound monitoring and surgical applications. *Advanced Healthcare Materials, 11*(10), 2200217. https://doi.org/10.1002/adhm.202200217